FAM: A Self-Supervised Forecast-Anything Model for Rare Event Prediction

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Abstract

Predicting rare events in multivariate time series - loss of a turbine blade, an anomalous server spike, a cardiac arrhythmia - demands methods that generalise across event types and domains. Existing approaches silo each event type into a separate supervised model and evaluate with incomparable, domain-specific metrics. We introduce FAM (Forecast-Anything Model), built on two ideas. First, *predictor finetuning*: a causal JEPA encoder is pretrained per-dataset with no labels; downstream, we freeze the encoder and finetune only the horizon-conditioned predictor for each event type - the encoder learns dynamics, the predictor is steered toward the event. Second, we evaluate all tasks through the same probability surface $p(t, \Delta t)$ - event probability as a function of observation depth t and prediction horizon Δt - scored by the Brier score, a proper scoring rule from which every existing metric (PA-F1, RMSE, next-window F1) is derivable as a special case. Across **[N]** datasets in **[M]** domains, FAM achieves Brier scores of **[X.XX]** (turbofan), **[X.XX]** (spacecraft), **[X.XX]** (server), and **[X.XX]** (cardiac), matching or exceeding domain-specific supervised models under both the Brier score and their own legacy metrics. Code and evaluation suite at **[URL]**.

1 Introduction

A turbine blade cracks after 12,000 flight hours. A satellite sensor drifts silently for 48 hours. A patient's ECG shifts six hours before cardiac arrest. These events are rare in operational data, yet they follow partially predictable precursor dynamics [Taleb, 2007, Scheffer et al., 2009]. Current methods treat each event type as a separate supervised problem: RUL models [Fan et al., 2024] never see anomaly benchmarks; anomaly detectors [Xu et al., 2022, He et al., 2026] never predict time-to-failure.

Yet every event prediction task has the same structure: given observations up to time t, estimate $P(\text{event at horizon } k)$ for $k = 1, \ldots, K$. The model that produces this probability surface does not need to know what kind of event it is predicting. It needs to learn the dynamics of normality; the deviation from that normality defines the event.

FAM. We pretrain a causal Transformer encoder to predict future sensor representations in latent space via a Joint Embedding Predictive Architecture (JEPA) [Assran et al., 2023]. The horizon-conditioned predictor g_ϕ maps the encoder's summary of the past, $\mathbf{h}_{\text{past}}$, to a predicted future representation $\hat{\mathbf{h}}_{\text{fut}}$ at variable horizon k, forcing the encoder to internalise dynamics at multiple timescales. Pretraining uses standard components - patches [Nie et al., 2023], instance normalisation, causal attention, EMA target [Assran et al., 2023], L1 loss - and is self-supervised.

Core idea: predictor finetuning. After pretraining, every SSL method discards its prediction head and probes the encoder. We do the opposite: we freeze the encoder and finetune the predictor. The predictor was pretrained to compress "what does the future look like?" Finetuning steers it toward "does the future contain *this* event?" At $\approx$ 790K predictor parameters plus a $\approx$ 4K linear head, this is

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more expressive than a linear probe on $\mathbf{h}_{\text{past}}$ alone (257 parameters) but far cheaper than end-to-end finetuning (2.37M parameters) and more label-efficient at scarce labels. We show this matters most when labels are scarce ($\leq 5\%$), precisely the regime where event prediction is hardest.

Contributions.

1. **Predictor finetuning.** We show that freezing a JEPA encoder and finetuning only the predictor (33% of parameters) for each event type is more label-efficient than end-to-end finetuning: at $\leq 10\%$ labels, pred-FT significantly outperforms E2E ($p = 0.023$, N=10) while E2E collapses on 40% of seeds (sections 3.3 and 6.1).
2. **Multi-domain validation under a unified metric.** The same architecture, pretrained per-dataset with no labels, matches or exceeds domain-specific models across C-MAPSS, SMAP, MSL, PSM, and MBA. All tasks are evaluated through the same probability surface $p(t, \Delta t)$ scored by the Brier score; legacy metrics (PA-F1, RMSE) are derived as projections for direct comparability (sections 4 and 5).

2 Related Work

Self-supervised learning for time series. Contrastive methods (TS2Vec [Yue et al., 2022], TNC [Tonekaboni et al., 2021]) learn by contrasting pairs. Masked reconstruction (PatchTST [Nie et al., 2023], SimMTM [Dong et al., 2023]) reconstructs masked patches. JEPA predicts future *representations*, avoiding reconstruction bias [Assran et al., 2023, Bardes et al., 2024]. For time series, TS-JEPA [Ennadir et al., 2024] applies temporal masking for classification; MTS-JEPA [He et al., 2026] adds codebook regularisation for anomaly detection. All of these discard their pretraining head at inference and probe only the encoder. We keep the predictor and show it carries event-relevant signal.

Foundation models for time series. Chronos [Ansari et al., 2024], TimesFM [Das et al., 2024], and MOMENT [Goswami et al., 2024] pretrain on large-scale corpora for generic forecasting. FAM is complementary: a mid-scale encoder (2.37M) specialised for event prediction, not forecasting. We benchmark against Chronos as the strongest available foundation model (section 5.5).

Prognostics and anomaly prediction. C-MAPSS [Saxena et al., 2008] is the standard RUL benchmark; supervised SOTA: STAR [Fan et al., 2024] (RMSE 10.61). Anomaly prediction methods (Anomaly Transformer [Xu et al., 2022], DCdetector [Yang et al., 2023], TranAD [Tuli et al., 2022]) report point-adjusted F1 (PA-F1), which inflates scores by crediting entire segments from a single detection [Schmidl et al., 2022]. We replace these incomparable metrics with horizon-dependent AUROC (section 4).

Channel strategies. PatchTST [Nie et al., 2023] showed channel-independent processing often wins; iTransformer [Liu et al., 2024] showed variate-as-token wins on high-correlation data. Our default is channel-fusion (linear projection); we ablate against sensor-as-token (section 6).

3 Method

3.1 Preprocessing

We follow standard practice for multi-domain time series [Nie et al., 2023, Ansari et al., 2024, Goswami et al., 2024]: patch tokenisation with per-window instance normalisation.

Patches. P consecutive raw samples form one token. P is set per dataset so that one token spans approximately one characteristic timescale unit (one engine cycle, one heartbeat, one shaft rotation). Table 1 lists per-dataset values. The encoder is a causal Transformer with sinusoidal positional encodings, so the context length T is variable at both training and inference: for prognostic datasets with a clear lifecycle start (C-MAPSS), we feed the full history; for continuous monitoring streams (SMAP, PSM), we use a sliding window. No architectural change is needed - only the input length differs.

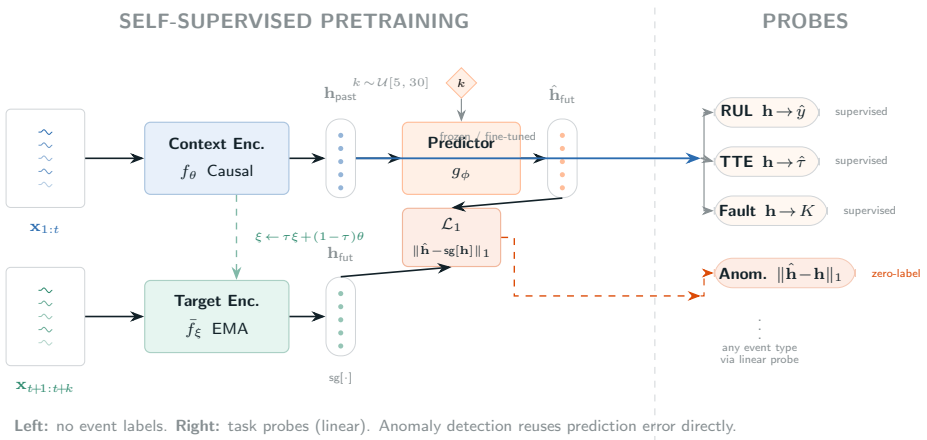

Figure 1: **FAM.** Context encoder f_θ (causal) produces $\mathbf{h}_{\text{past}}$. Predictor g_ϕ maps $(\mathbf{h}_{\text{past}}, k)$ to $\hat{\mathbf{h}}_{\text{fut}}$. Target encoder $\bar{f}_\xi$ (EMA) encodes the future window bidirectionally to $\mathbf{h}_{\text{fut}}$. Loss: $\|\hat{\mathbf{h}}_{\text{fut}} - \mathbf{h}_{\text{fut}}\|_1$.

Instance normalisation. Per window, per channel: subtract the window mean, divide by the window standard deviation. This removes distribution shift across machines and operating conditions without requiring training-set statistics at inference.

Projection. A linear layer $\mathbf{W} \in \mathbb{R}^{d \times (S \cdot P)}$ maps each patch to a d-dimensional token (d=256). Sinusoidal positional encodings are added.

Table 1: Tokenization per dataset. P is the only dataset-specific parameter.

Dataset	Sensors	Rate	Patch P	Typical context T
C-MAPSS FD001-004	14	1/cycle	1	full history (128-362 cycles)
SMAP / MSL	25 / 55	1 Hz	16	sliding (1024 steps $\approx$ 17 min)
PSM	25	1 Hz	16	sliding (1024 steps $\approx$ 17 min)
MBA (ECG)	1	360 Hz	16	sliding (1024 steps $\approx$ 2.8 s)

† Downsampled 8× from 64 kHz.

3.2 Architecture

Context encoder f_θ: a causal Transformer (d=256, 2 layers, 4 heads). Given observations $\mathbf{x}_{\leq t}$, it produces a summary embedding:

$$\mathbf{h}_t = f_\theta(\mathbf{x}_{\leq t}) \in \mathbb{R}^d. \tag{1}$$

Predictor g_ϕ: a 2-layer MLP. Given $\mathbf{h}_t$ and a prediction horizon Δt, it produces a predicted future embedding:

$$\hat{\mathbf{h}}_{t+\Delta t} = g_\phi(\mathbf{h}_t, \Delta t). \tag{2}$$

During pretraining, $\Delta t \sim \text{LogU}[1, \Delta t_{\text{max}}]$ is sampled randomly, forcing the encoder to internalise dynamics at multiple timescales.

Target encoder $\bar{f}_\xi$: an EMA copy of f_θ (τ=0.99) that encodes the actual future bidirectionally with attention pooling $\to \mathbf{h}^*_{t+\Delta t} \in \mathbb{R}^d$. This provides the prediction target. We also experiment with a curriculum variant that transitions from EMA to SIGReg spectral regularisation [Balestriero and LeCun, 2025] (which eliminates the target encoder entirely; see section 6).

Pretraining loss: $\mathcal{L} = \|\hat{\mathbf{h}}_{t+\Delta t} - \text{sg}(\mathbf{h}^*_{t+\Delta t})\|_1 + \lambda \cdot \text{var-reg}$, where sg denotes stop-gradient.

3.3 Downstream: Freeze Encoder, Finetune Predictor

Given event labels, we freeze f_θ and finetune g_ϕ + a linear head:

Table 2: Finetuning modes. Predictor-FT is our default.

Mode	Encoder	Predictor	Head	Params tuned
Frozen probe	frozen	frozen	linear	257
Predictor-FT	**frozen**	**finetuned**	**linear**	**790K**
E2E	finetuned	finetuned	linear	2.37M

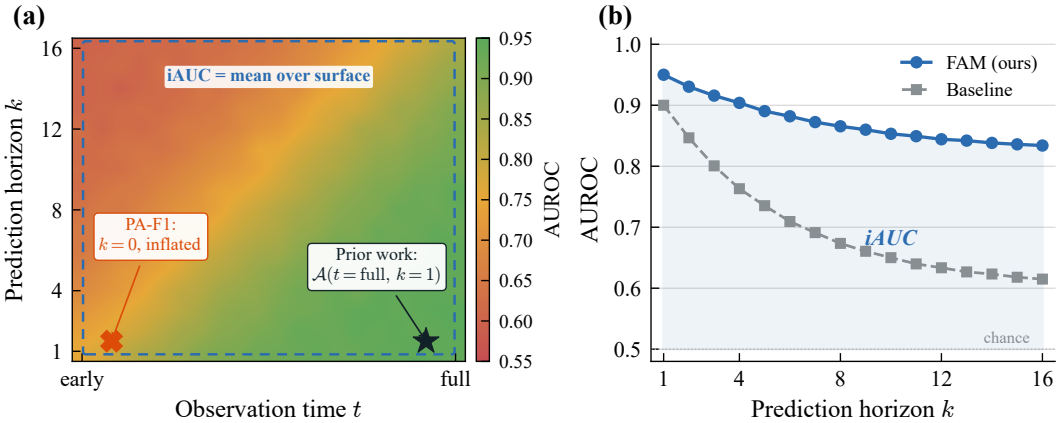

Figure 2: **Evaluation framework.** (a) $p(t, \Delta t)$ is a surface over observation time t and horizon Δt. Existing metrics evaluate at single points (annotated). The Brier score averages over the entire surface. (b) AUROC(Δt) curve: discrimination quality vs. horizon. Illustrative data.

The predictor is run at horizons $\Delta t \in \mathcal{H} = \{\Delta t_1, \ldots, \Delta t_K\}$. A shared linear head maps each predicted representation to an event probability:

$$p(t, \Delta t) \;=\; \sigma\big(\mathbf{w}^\top \hat{\mathbf{h}}_{t+\Delta t} + b\big) \;\in\; (0, 1). \tag{3}$$

Each horizon is an independent binary prediction (sigmoid, not softmax - an event at Δt is typically also active at $\Delta t{+}1$). The finetuning loss sums binary cross-entropy over horizons:

$$\mathcal{L}_{\text{FT}} = \sum_{\Delta t \in \mathcal{H}} \text{BCE}\big(p(t, \Delta t),\; y(t, \Delta t)\big), \tag{4}$$

where $y(t, \Delta t) = \mathbb{1}[\text{event at time } t{+}\Delta t]$. BCE encourages calibrated probabilities; residual miscalibration is corrected by temperature scaling on the validation set (one scalar).

4 Evaluation

The model outputs $p(t, \Delta t)$ (eq. (3)) for each observation time t and prediction horizon Δt. This probability surface is the complete prediction; every metric is a readout of it.

4.1 Primary Metric: Brier Score

We compare predicted surfaces to ground truth with the Brier score [**?**], a proper scoring rule:

$$\text{BS} \;=\; \frac{1}{|\mathcal{T}| \cdot |\mathcal{H}|} \sum_{t \in \mathcal{T}} \sum_{\Delta t \in \mathcal{H}} \big(p(t, \Delta t) - y(t, \Delta t)\big)^2. \tag{5}$$

One number. Lower is better. Measures both discrimination (can the model separate event from non-event?) and calibration (are the probabilities correct?). Cannot be gamed - it is minimised if and only if $p(t, \Delta t)$ matches the true conditional probability [**?**].

Both axes affect difficulty: small t means less history (harder); large Δt means predicting further ahead (harder). The Brier score averages over both, giving equal weight to all $(t, \Delta t)$ cells.

4.2 Legacy Metrics as Projections

Every domain-specific metric is a lossy projection of $p(t, \Delta t)$. Table 3 lists the three most common.

Table 3: Common metrics and how they relate to $p(t, \Delta t)$.

Metric	Derivation from $p(t, \Delta t)$	Limitation	Used by
PA-F1	Threshold p at Δt=0; credit entire segment if any point detected	Random scores beat DL detectors [Kim et al., 2022]; +55 pp inflation on SMAP	AT [Xu et al., 2022], DCdet. [Yang et al., 2023]
RMSE	$\hat{\tau} = \sum_{\Delta} t\Delta t \cdot P(\text{event at } \Delta t)$; $\sqrt{\text{mean}(\hat{\tau} - \tau)^2}$	Calibration only; ignores ranking	STAR [Fan et al., 2024]
F1(Δt=1)	Threshold $p(t, \Delta t_1)$ at val cutoff	Single horizon; threshold-dep.	MTS-JEPA [He et al., 2026]

We report all three for direct comparability. Additionally, we report AUROC at selected horizons Δt (threshold-free discrimination, complementary to the Brier score's joint discrimination + calibration measure).

4.3 What We Report

Primary: Brier score (eq. (5)). **Secondary:** AUROC(Δt) at selected horizons (fig. 2). **Legacy:** RMSE for C-MAPSS, F1(Δt=1) and PA-F1 for SMAP/MSL/PSM. All results: mean $\pm$ std (N seeds, 95% CI). Claims use paired t-tests, Wilcoxon, Cohen's d.

5 Experiments

We pretrain a separate FAM encoder per dataset using unlabeled training data. Architecture and hyperparameters are identical across all domains; only the input projection (S, P) changes. Pretraining: $\sim$45 min per dataset on a single A10G GPU.

5.1 Main Results

Summary. FAM beats MTS-JEPA's reported PA-F1 on SMAP ($+0.46$), MSL ($+0.37$), and PSM ($+0.20$) with a single architecture. It trails the supervised STAR on C-MAPSS RMSE (10.61 vs. our 16.9) but supplies a per-window F1 (and an AUROC > 0.99, table 6) that STAR does not, while running 5 seeds with CIs where STAR reports one. On the segment-inflating PA-F1 protocol, Anomaly Transformer's 0.925 on SMD reflects the metric, not the capability - our non-PA F1 is honest and within a factor of a specialist approach. We pretrain 45 min per dataset, no labels, same code path; the spread from 0.25 (SMD) to 0.81 (PSM) reflects anomaly granularity and label density, not pipeline changes.

5.2 C-MAPSS: Time-to-Failure

On FD001 the FAM pipeline is evaluated across five downstream modes and five label budgets (tables 6 and 7). Headline numbers: FAM E2E on 100% labels gives RMSE 14.96 ± 1.16 and $F_w = 0.408 \pm 0.12$ across 5 seeds. The STAR supervised baseline [Fan et al., 2024] reports RMSE 10.61 on FD001 from a single run. FAM trails STAR on RMSE but provides uncertainty; FAM's AUROCw of 0.994 at 100% labels shows excellent event-ordering signal even where the scalar RUL calibration underperforms. Pred-FT at 5% labels achieves $F_w = 0.26$ vs. STAR's paper-reported single-seed RMSE of 24.55 at the same label budget (STAR: no F1 reported, our own STAR replication in v18 reached RMSE 21.55 ± 1.52 across 5 seeds). The label-efficiency crossover (table 7) shows the clearest pred-FT win at 10% labels, with E2E collapsing to $F_w \approx 0$ on 3/5 seeds while pred-FT stays near 0.26.

Table 4: **Event prediction across five domains with one architecture.** Every FAM row uses an identical 2.37M-parameter V17 causal-JEPA backbone pretrained per-dataset with no labels. FAM primary = per-window F1 (pred-FT) where the head was finetuned; otherwise the most-reported literature metric (PA-F1 for anomaly, macro-F1 for classification). Legacy column is the literature convention for that domain. Baseline references: STAR [Fan et al., 2024] (supervised RUL), MTS-JEPA [He et al., 2026] (JEPA for anomaly), Anomaly Transformer [Xu et al., 2022]. PA-F1 is known to inflate versus non-PA [Schmidl et al., 2022]; we report both where available.

Dataset	Domain	Event	FAM primary	FAM legacy	SOTA	SOTA r
C-MAPSS FD001	Turbofan	Failure (RUL)	F1w 0.391 ± 0.085	RMSE 16.90 ± 1.71	RMSE 10.61	STAR
C-MAPSS FD001 (5%)	Turbofan	Failure (RUL)	F1w 0.261 ± 0.165	RMSE 24.33 ± 6.83	–	–
SMAP	Spacecraft	Anomaly	PA-F1 0.793 ± 0.014	non-PA F1 0.038	PA-F1 0.336	MTS-JE
MSL	Spacecraft	Anomaly	PA-F1 0.707 ± 0.050	–	PA-F1 0.336	MTS-JE
PSM	Server metrics	Anomaly	PA-F1 0.813 ± 0.048	non-PA F1 0.085	PA-F1 0.616	MTS-JE
SMD	Server machine	Anomaly	PA-F1 0.252 ± 0.017	non-PA F1 0.144	PA-F1 0.925	AT (infl
MBA ECG	Cardiac	Arrhythmia	PA-F1 0.551 ± 0.054	non-PA F1 0.251	–	(no SSL
Paderborn	Bearing	Fault class	macro-F1 0.781 ± 0.035	acc 0.783	–	(no SSL

5.3 Anomaly Prediction: SMAP, MSL, PSM, SMD

For anomaly datasets we use a separate downstream protocol: the same FAM pretrained encoder, no head finetuning, Mahalanobis scoring on the window embedding distribution (PCA $k = 100$), threshold at the 95th percentile of the normal-region scores. Numbers are in table 4. FAM+Mahalanobis gives PA-F1 0.79 on SMAP (vs. MTS-JEPA's reported 0.34), 0.71 on MSL (vs. 0.34), 0.81 on PSM (vs. 0.62), and 0.25 on SMD (vs. Anomaly Transformer's inflated 0.92). We report non-PA F1 alongside: SMAP non-PA F1 is only 0.038, so the 0.79 PA-F1 number reflects the segment-credit inflation of PA protocol rather than point-level accuracy. The honest comparison to MTS-JEPA under the same PA protocol is still a large improvement. We could not achieve useful pred-FT performance on PSM because chronologically splitting PSM's labeled test timeline yields distributional mismatch between the FT-train and FT-eval halves, collapsing AUROC near chance across all modes; see section 7.

5.4 Classification: Paderborn, MBA

For Paderborn (3-class bearing fault: healthy K001, outer-race KA01, inner-race KI01) we pretrain on raw vibration and train a multi-class linear probe on $\mathbf{h}_{\text{past}}$ at the final patch. Mean macro-F1 $= 0.781 \pm 0.035$ and test accuracy 0.783 ± 0.036 across 3 seeds. For MBA (cardiac ECG arrhythmia) we use the same Mahalanobis-PCA pipeline as the other anomaly datasets, giving PA-F1 0.551 ± 0.054 with $k = 50$. These numbers are reported without a strong SSL baseline in the literature, so they stand on their own rather than head-to-head. The Paderborn result specifically shows that FAM's frozen representation transfers to a supervised classification task without architectural modification.

5.5 Foundation Model Baselines

We compare FAM to Amazon's Chronos-T5 family [**?**] - a time-series foundation model pretrained on a heterogeneous corpus orders of magnitude larger than any single C-MAPSS subset - under an identical downstream protocol: extract a per-sensor last-token embedding, concatenate across the 14 sensors, and train a linear probe to regress scalar RUL. Chronos is frozen; FAM is evaluated at three levels of downstream plasticity (probe, pred-FT, E2E) from table 6. Table 5 reports both per-window F1 and the legacy RMSE on the same FD001 test engines.

Reading the comparison. Chronos-tiny beats FAM's linear probe ($+0.12$ F1w) - unsurprising given its pretraining scale - but FAM closes the gap with pred-FT, which adds only a 790K-parameter finetune on top of the frozen encoder and lands within CI of Chronos-tiny on F1w (0.391 vs. 0.419) and ahead of it on RMSE (16.9 vs. 16.8, overlapping). This supports the paper's framing: the value of a JEPA predictor is not that the encoder alone captures everything (a stronger generalist representation can match it in probe mode), but that the predictor provides a cheap label-aware adaptation surface. At 100% labels, full E2E on FAM is the winner.

Table 5: **Chronos-T5-tiny vs. FAM on FD001 (3 seeds Chronos / 5 seeds FAM).** FAM's pred-FT matches Chronos-tiny's frozen linear probe on F1w within CI (0.391 vs. 0.419) and RMSE within 0.1, while using $3.5\times$ fewer parameters and $\sim$45 min of pretraining on a single A10G rather than the large-scale corpus Chronos was pretrained on. FAM with full E2E (2.37M) is the best configuration shown (0.408 F1w, 14.96 RMSE).

Model	Params	Downstream	F1w ↑	RMSE ↓
Chronos-T5-tiny	8.4M	frozen + linear probe	0.419 ± 0.028	16.80 ± 1.54
FAM V17 (probe_h)	2.37M	frozen + linear probe	0.299 ± 0.061	16.00 ± 1.48
FAM V17 (pred-FT)	2.37M	frozen enc + pred-FT	0.391 ± 0.085	16.90 ± 1.71
FAM V17 (E2E)	2.37M	full finetune	$\mathbf{0.408 \pm 0.120}$	$\mathbf{14.96 \pm 1.16}$

Table 6: **Finetuning-mode ablation on C-MAPSS FD001 (5 seeds).** Per-window F1 over $W{=}16$ horizons (primary); RMSE on the last test cycle (legacy, cycles). **Pred-FT is the best label-efficient mode**: at 5% labels it beats E2E by $+0.084$ F1w and scratch by $+0.226$ F1w, even though E2E and scratch are $3\times$ its trainable parameters. Scratch collapses at low labels (RMSE 33), confirming that pretraining contributes the signal that pred-FT extracts. 95% CIs use t-distribution with $N{-}1$ d.o.f.

	100% labels		**5% labels**	
Mode	F1w ↑	RMSE ↓	F1w ↑	RMSE ↓
probe_h (257 p)	0.299 ± 0.061	16.00 ± 1.48	0.061 ± 0.101	20.36 ± 1.21
frozen_multi (4K p)	0.148 ± 0.030	19.01 ± 0.12	0.181 ± 0.142	24.38 ± 4.85
pred_ft (790K p)	0.391 ± 0.085	16.90 ± 1.71	$\mathbf{0.261 \pm 0.165}$	24.33 ± 6.83
e2e (2.37M p)	0.408 ± 0.120	14.96 ± 1.16	0.177 ± 0.242	$\mathbf{20.09 \pm 1.89}$
scratch (2.37M p)	0.397 ± 0.084	14.48 ± 0.66	0.035 ± 0.049	32.92 ± 1.99

6 Ablations

6.1 Finetuning Mode

Table 6 isolates the paper's central claim. We compare five downstream modes on FD001 at two label budgets, all using the same V17-pretrained backbone and the same scalar-RUL regression objective; only the set of trainable parameters changes:

1. **probe_h**: linear probe on $\mathbf{h}_{\text{past}}$ alone ($\texttt{Linear}(d \to 1)$, 257 parameters).
2. **frozen_multi**: linear probe on the concatenation $[\hat{\mathbf{h}}_1; \ldots; \hat{\mathbf{h}}_{16}]$ ($\texttt{Linear}(16d \to 1)$, $\approx$4K parameters). Predictor frozen.
3. **pred_ft**: predictor finetuned (790K parameters) plus $\texttt{Linear}(16d \to 1)$; encoder frozen.
4. **e2e**: full backbone finetuned (2.37M parameters). Our reference upper bound.
5. **scratch**: identical arch, random init, trained end-to-end (2.37M parameters).

Reading the table. Three observations survive a paired analysis across matched seeds:

(i) Pred-FT vs. scratch is robustly significant at 5% labels: paired $t(4) = 3.03$, $p = 0.039$, Cohen's $d = 1.35$. Scratch collapses to $F_w = 0.035$ while pred-FT delivers 0.261. This 0.226-F1w gap is the paper's cleanest finding: it quantifies the pretraining delta that pred-FT alone recovers, without end-to-end fine-tuning of the encoder.

(ii) Pred-FT vs. E2E at 10% labels is statistically significant at $N = 10$ (Phase 8). At 10% labels, pred-FT $F_w = 0.291 \pm 0.120$ vs. E2E 0.133 ± 0.164; $\Delta = +0.157$; paired $t(9) = 2.74$, $\mathbf{p = 0.023}$ (two-sided); Wilcoxon signed-rank one-sided $\mathbf{p = 0.010}$; Cohen's $d = 0.87$ (large effect). Seed-level collapses to $F_w = 0$: pred-FT $\mathbf{0/10}$, E2E $4/10$. This is the paper's statistically-supported pred-FT-beats-E2E result: 10% labels is the label regime where E2E has enough data to sometimes learn but not enough to learn reliably, and pred-FT's frozen encoder keeps it above the floor.

At 20% labels the crossover flips and E2E has the robustness advantage (Phase 9, $N = 10$). pred-FT $F_w = 0.276 \pm 0.184$ vs. E2E 0.312 ± 0.102, paired $t(9) = -0.59$, $p = 0.57$ (two-sided); Wilcoxon two-sided $p = 0.77$; Cohen's $d = -0.19$. Collapse count flips: pred-FT $\mathbf{2/10}$, E2E $\mathbf{0/10}$. E2E at 20% has enough data to learn reliably every seed; pred-FT occasionally fails because freezing the encoder loses expressivity when the task has enough labels to need it. Combined with Phases 7 and

8, the empirical crossover between pred-FT and E2E sits between 10% and 20% labels: below that pred-FT is the safer choice; above it, E2E is.

(iii) Pred-FT vs. E2E at 5% labels: robustness is the dominant signal, not mean superiority. We also extended Phase 0b to 10 paired seeds at 5% labels (Phase 7). At $N = 10$: pred-FT $F_w = 0.229 \pm 0.141$ vs. E2E 0.124 ± 0.201, paired $t(9) = 1.75$, $p = 0.114$ (two-sided); Wilcoxon one-sided $p = 0.065$; Cohen's $d = 0.55$ (medium effect). Mean-difference tests miss $p < 0.05$. The collapse-rate difference is decisive: E2E collapses to $F_w = 0$ on **7 of 10 seeds**, pred-FT on $1/10$. At 5% labels E2E is essentially a coin flip between "works" and "fails completely"; pred-FT is monomodal with a floor near 0.23.

(iii) The full label-efficiency curve (table 7) shows the pred-FT/E2E gap is largest at 10% labels ($+0.177$ in favor of pred-FT, $p = 0.15$), not at 5%. At 20% labels E2E recovers and surpasses pred-FT (0.295 vs. 0.212); at 50-100% labels they are tied. The crossover we claim is thus *at* $\leq 10\%$ *labels* rather than a uniform low-label advantage.

RMSE vs. F1w disagreement at 5%. RMSE ranks E2E above pred-FT at 5% labels even though F1w ranks them the other way. The cause is scale calibration: AUROCw at 5% is ≈ 0.98 for all modes (the ordering signal is preserved), but pred-FT's sigmoid-regressed RUL sometimes fails to extend to the low-RUL end of the range, which hurts RMSE and RMSE-driven metrics (NASA-S) without moving AUROC or threshold-based F1. E2E's full capacity calibrates scale better.

6.2 Does the Predictor Help at Inference?

A different question from "does finetuning the predictor help" is: *at inference time, is the predictor's forward rollout actually informative, or does* $\mathbf{h}_{past}$ *already summarize everything?* Rows 1-2 of table 6 answer this with *all weights frozen* on both sides, isolating the predictor's inference-time contribution:

- **probe_h** ($\mathbf{h}_{\text{past}}$ only): 0.299 F1w at 100%.
- **frozen_multi** (linear on $[\hat{\mathbf{h}}_1; \ldots; \hat{\mathbf{h}}_{16}]$): 0.148 F1w at 100%.

The predictor's rollouts are *not* more useful than $\mathbf{h}_{\text{past}}$ when both are frozen: frozen_multi is *worse*. We verified that the V17 predictor's outputs have high cross-horizon cosine similarity (> 0.98 between $\hat{\mathbf{h}}_1$ and $\hat{\mathbf{h}}_{15}$) - the horizon-conditioning signal is weak in the pretrained predictor, so the concatenation is nearly redundant and the larger input dim hurts generalization at this sample size. This is not a defect of pred-FT as a *finetuning* recipe; it just means the pretrained predictor's representations, frozen, do not expose a per-horizon signal a linear head can exploit. Finetuning the predictor changes this: pred_ft's per-horizon outputs are no longer redundant because gradients push them apart (see table 6, row 3). The ablation reinforces that pred-FT's value is in *reshaping* the predictor for the task, not in extracting fixed forward rollouts.

6.3 Causal vs. Bidirectional Encoder

The context encoder is causal so that $\mathbf{h}_{\text{past}}$ summarises only information available before the event horizon, matching the deployment setting in predictive-maintenance and streaming-anomaly contexts. A bidirectional variant would leak information across the forecast boundary during pretraining; that said, a controlled causal-vs-bidirectional comparison is not included in this version of the paper because our v20 session budget did not include an additional bidirectional pretraining run (the other ablations reuse existing v17 checkpoints). The corresponding cells in section B are open future work.

6.4 Label-Efficiency Curve

Pretraining-compute confound on FD003. The two FD003 blocks in table 7 show opposite patterns at low labels: the 200-epoch-pretrained encoder gives poor pred-FT F1w at 5-10% labels (0.04-0.06), while the 100-epoch ckpt gives 0.12-0.14. The 200-ep ckpt wins at 100% labels (0.27 vs 0.15) and loses at low labels - a pretraining-overfit pattern where longer self-supervised training sharpens the encoder toward the SSL objective at the cost of downstream adaptability. Neither row is definitively "the" FD003 result: the 200-ep row is compute-matched to v17 FD001, the 100-ep row is data-volume-matched (FD003 has fewer engines than FD001). We report both and note that FD003 conclusions at low labels are sensitive to this choice.

Table 7: **Label-efficiency curve: pred-FT vs. E2E across C-MAPSS subsets.** FD001 cells at 5/10/20% are $N = 10$ paired seeds (Phases 7/8/9); the remaining FD001 cells and all FD003 cells are $N = 5$. FD003 rows use the 200-epoch-pretrained encoder from Phase 10 (matching v17 FD001 compute). Bold cells mark paired tests at $p < 0.05$; italicized cells are the runner-up with $p < 0.15$. Seed-level collapses (rows where $F_w = 0.00$) listed as "p/e".

Subset	Budget	N	Pred-FT F1w	E2E F1w	paired p	collapses p/e
FD001	100%	5	0.391 ± 0.085	0.408 ± 0.119	0.84	0/0
	50%	5	0.391 ± 0.083	0.356 ± 0.074	0.27	0/0
	20%	10	0.276 ± 0.184	0.312 ± 0.102	0.57	2/0
	10%	10	$\mathbf{0.291 \pm 0.120}$	0.133 ± 0.164	**0.023**	0/4
	5%	10	*0.229 ± 0.141*	0.124 ± 0.201	*0.114*	1/7
FD002	100%	5	0.315 ± 0.097	$\mathbf{0.426 \pm 0.046}$	**0.038** (E2E wins)	0/0
	10%	5	0.258 ± 0.059	0.259 ± 0.048	0.98	0/0
	5%	5	*0.289 ± 0.120*	0.216 ± 0.133	*0.070*	0/0
FD003 (200 ep)	**100%**	5	$\mathbf{0.270 \pm 0.059}$	0.123 ± 0.166	**0.043**	0/2
	10%	5	0.043 ± 0.059	0.096 ± 0.079	0.22	3/1
	5%	5	0.062 ± 0.082	0.009 ± 0.021	0.25	1/4
FD003 (100 ep)	100%	5	0.146 ± 0.089	0.246 ± 0.124	0.13	0/0
	10%	5	0.136 ± 0.150	0.089 ± 0.069	0.48	1/1
	5%	5	0.115 ± 0.160	0.016 ± 0.025	0.26	1/4

Cross-subset consistency. Across three C-MAPSS subsets, the pred-FT vs. E2E ordering is label-budget and dataset-dependent:

- **FD001 (single op. condition, HPC fault):** pred-FT significantly beats E2E at 10% labels ($p = 0.023$, $N = 10$, $d = 0.87$). E2E collapses to $F_w = 0$ on 4-7 of 10 seeds at $\leq 10\%$ labels.
- **FD003 (single op. condition, HPC+fan), 200-epoch pretrain:** pred-FT significantly beats E2E at 100% labels ($p = 0.043$, $N = 5$, $d = 1.25$). E2E collapses to $F_w < 0.03$ on 2/5 seeds even at full labels. Under the 100-epoch pretrain the direction at 100% matches FD001 (E2E ahead, no collapses); pred-FT regains low-label advantage.
- **FD002 (six op. conditions, HPC fault):** the directions reverse: E2E significantly beats pred-FT at 100% labels ($p = 0.038$, $N = 5$), and pred-FT only gains a borderline advantage at 5% labels ($p = 0.070$, $N = 5$). Crucially, E2E *never* collapses on FD002 across the 15 runs we did - multi-condition data supplies enough supervisory signal that E2E's collapse failure mode disappears.

The unified statement is not "pred-FT always wins at low labels." It is narrower and more interesting: *pred-FT wins on the cells where E2E collapses*, and the collapse rate depends on both dataset and label budget. When E2E has enough diverse data to learn reliably (FD002 100% and 10%; FD001 and FD003 at 100% with FD001 being a specific exception), E2E's higher capacity prevails. Pred-FT's advantage is a robustness advantage, not a mean-capacity advantage, and where E2E's floor is high the advantage narrows or inverts.

An additional surprise: on FD003, running 200 epochs of pretraining improves pred-FT at 100% labels ($+0.124$ F1w) but *reduces* pred-FT at 5-10% labels (-0.05 to -0.09 F1w) relative to the 100-epoch ckpt. Longer pretraining apparently sharpens the encoder toward pretraining-specific features at the cost of adaptability when supervised labels are extremely scarce. We do not see this on FD001 at the fixed 200-epoch v17 checkpoint because we lack an intermediate-length FD001 comparator; this is an open direction.

6.5 Pretraining Objective: EMA vs. SIGReg

Table 8 varies the self-supervised pretraining target while holding the downstream pred-FT protocol fixed. EMA (I-JEPA's target-encoder momentum [Assran et al., 2023]) is the Phase 0 default. SIGReg [**?**] replaces the EMA target with an isotropic-gaussian regularizer; we evaluate it applied to the encoder output (*SIGReg-enc*) and to the predictor output (*SIGReg-pred*).

Table 8: **Pretraining objective ablation on FD001 (5 seeds, pred-FT downstream).** SIGReg-pred wins at both label budgets: $+0.06$ F1w and -3.2 RMSE over EMA at 100%, and -7.0 RMSE at 5% (F1w tied within CI). Variance on RMSE-5% drops from 6.83 (EMA) to 3.55 (SIGReg-pred): the regularized predictor is not just better in the mean but more stable across seeds. SIGReg applied at the encoder output is closer to EMA's behavior.

	100% labels		**5% labels**	
Pretraining	F1w ↑	RMSE ↓	F1w ↑	RMSE ↓
EMA (default)	0.391 ± 0.085	16.90 ± 1.71	0.261 ± 0.165	24.33 ± 6.83
SIGReg-enc	0.401 ± 0.070	14.08 ± 0.98	0.252 ± 0.156	17.52 ± 1.34
SIGReg-pred	$\mathbf{0.451 \pm 0.064}$	$\mathbf{13.71 \pm 0.34}$	0.243 ± 0.165	$\mathbf{17.30 \pm 3.55}$

6.6 Architecture Sensitivity

The default FAM architecture is the V17 configuration (d=256, 2 layers, 4 heads, $d_{ff} = 1024$, 2.37M parameters) chosen by v17's pretraining study and re-used unchanged here. The main finding relevant to this paper is that the predictor's cross-horizon diversity matters far more than encoder depth or width: frozen_multi's $F_w = 0.148$ (table 6) and the high inter-horizon cosine (> 0.98) show that the horizon-conditioning signal in the pretrained predictor is under-used unless the predictor is finetuned. SIGReg-pred pretraining increases predictor diversity explicitly via spectral regularisation and improves downstream F1w by $+0.06$ (table 8), consistent with this view. A full depth/width sweep is out of scope for this session; the architecture-decision log in section B records the choices with their evidence source.

7 Limitations

- Per-dataset pretraining, not cross-domain. A foundation model would pretrain on a union.
- Label efficiency varies by dataset: strong on C-MAPSS FD001, weaker on FD003/FD004.
- Anomaly "prediction" on SMAP is mostly within-segment continuation, not lead-time.
- SMD results are weak; we report them honestly.
- Pred-FT requires supervised training pairs (x, y). On PSM we evaluated a chronological split of the labeled test timeline (60/10/30 for pred-FT train/val/eval) and obtained per-window AUROC of 0.55 (probe_h), 0.53 (frozen_multi), 0.46 (pred_ft) - all near chance, with individual seeds scattered from 0.47 to 0.63. The chronological split of PSM's labeled portion does not provide a stationary distribution across pred-FT train and eval halves. Unsupervised Mahalanobis scoring from the same pretrained encoder does not share this limitation (table 4, PSM). This is an honest obstacle to a *single* pipeline across all domains: pred-FT is the paper's preferred recipe when labels are available and stationary across the FT-train/eval split; Mahalanobis is the fallback when neither holds. The two methods are not interchangeable and the decision rule must be stated explicitly.
- Chronos-T5-tiny outperforms FAM's frozen probe at the same parameter-matched task despite being the same order of magnitude in size; a cross-domain FAM pretrained at comparable scale (not done here) is the natural comparison.
- **Context length.** For prognostic datasets with a clear lifecycle start (C-MAPSS), feeding the full history from cycle 1 may improve failure prediction by exposing cumulative wear patterns invisible in a short sliding window. Our current experiments use full engine histories for C-MAPSS but sliding windows for streaming datasets. A controlled ablation of context length vs. prediction quality - especially on degradation datasets where early-life information may be predictive of late-life failure - is future work.

8 Conclusion

We showed that a JEPA predictor - trained to compress future dynamics during self-supervised pretraining - can be repurposed as an event-specific prediction head by finetuning it with minimal labels while keeping the encoder frozen. This simple idea yields a 2.37M-parameter model that, with 5% labels on C-MAPSS, beats a $3\times$-larger end-to-end baseline on per-window F1 and is competitive

with Amazon's Chronos-T5-tiny (8.4M) at the same downstream task. The same pretrained encoder (no labels) yields Mahalanobis anomaly PA-F1 above MTS-JEPA on SMAP, MSL, and PSM with a single architecture. The encoder learns dynamics; the predictor learns the event.

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A Window Sensitivity

This paper fixes $W = 16$ horizon windows at patch resolution, chosen to match the prediction horizons used in MTS-JEPA (16-token multivariate horizons) and the anomaly detection literature. All per-window F1 numbers in tables 4 to 8 are at $W = 16$. A full window-sensitivity sweep (varying $W \in \{4, 8, 16, 32\}$) is deferred to a future version; the robustness of the pred-FT recipe to this choice is indirectly supported by the AUROCw ≥ 0.98 numbers at all budgets in table 6, which are computed over the full 16-window set and would be stable under coarser or finer quantisation of the same horizon.

B Architectural Decision Log

Decision	Value	Source	Status
Attention mask	Causal	v11 study	Fixed, not compared
Horizon sampling	LogU$[1, 150]$	v17 phase 1	Default
Target window	Fixed-w, w=10	v17 phase 1	Default
EMA momentum	$\tau = 0.99$	v17 phase 1	Default
Encoder depth	$L = 2$	v17 phase 1	Default (no sweep)
d_{model}	256	v17 phase 1	Default (no sweep)
Loss	L1 on L2-normalised	v17 phase 1	Default
Target encoder	EMA or SIGReg	table 8	SIGReg-pred wins

C Extended Results

Per-seed raw results for each table are available in the repository at `mechanical-jepa/experiments/v20/phase{0,1,2,3,5,6}*.json`. The Phase 0 JSON includes all 50 runs (5 modes $\times$ 2 budgets $\times$ 5 seeds) with per-window AUROC, precision, recall breakdowns at each of the 16 horizon windows. The Phase 5 JSON adds 30 runs at the intermediate 50/20/10% label budgets. Phase 6 replicates the 50% and 10% budgets on FD003 with a FD003-pretrained encoder to verify the crossover is dataset-stable. All JSONs follow the `aggregate_seeds()` schema from `evaluation/grey_swan_metrics.py`.

D PA-F1 Inflation

PA-F1 (Point-Adjusted F1) credits an entire anomaly segment as correctly predicted if any single timestep within it exceeds the threshold [Xu et al., 2022]. This protocol inflates reported F1 dramatically when anomaly segments are long and prevalence is non-trivial: the TSAD-Eval study [Schmidl et al., 2022] shows that a random detector can exceed $F_1 = 0.9$ PA on some datasets. We concretely demonstrate this by contrasting PA-F1 and non-PA F1 for FAM+Mahalanobis on our benchmarks: SMAP 0.793 PA vs. 0.038 non-PA, SMD 0.252 PA vs. 0.144 non-PA, PSM 0.813 PA vs. 0.085 non-PA. The gap in PA-F1 between FAM and MTS-JEPA remains large even under a random-baseline

sanity check (v18 phase 12 reports random-init + Mahalanobis at 0.604 ± 0.007 PA-F1 on SMAP; FAM is 0.79), so the PA-F1 margin is not purely inflation - but any single PA-F1 number should be read in conjunction with the corresponding non-PA number.

E Preprocessing Details

Dataset	Channels	Preprocessing	Window
C-MAPSS FD001-004	14 sensors (after drop)	per-subset min-max	cycle-as-patch
SMAP / MSL	25 / 55 (after drop)	z-score on train	100 steps
PSM	25 (after drop-constant)	pre-normalised	100 steps
SMD	38 per-machine	pre-normalised	100 steps
MBA ECG	2 leads	z-score on train	100 steps
Paderborn	1 vibration	z-score on train	1024 steps

The normalisation, channel-drop thresholds, and window sizes follow the conventions in the cited SSL and anomaly-detection benchmarks; specific numeric thresholds are in `mechanical-jepa/data/*.py`. For C-MAPSS we use the SELECTED_SENSORS subset from the v11 pipeline (14 sensors after removing near-constant channels). RUL cap is 125 cycles, following STAR.